Great idea — this sits nicely at the intersection of **personalized education, LLMs, and user modeling**. Here is a full project proposal in your lab's format:

Project Proposal: EduTwin — LLM-Powered Digital Twin of University Students

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1. Motivation & Problem Statement

University students differ vastly in learning pace, prior knowledge, preferred study style, and academic trajectory. Yet most educational systems treat them uniformly. A **digital twin** — a dynamic, data-driven model of an individual student — can enable personalized interventions, early dropout prediction, adaptive content delivery, and intelligent tutoring.

LLMs make this tractable for undergraduates: instead of building complex probabilistic user models from scratch, students can represent a learner's state as a **structured profile** and use LLM reasoning over that profile to simulate behavior, answer questions, and generate personalized content.

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2. Core Idea

Build a text-based digital twin of a student by constructing a **Live Learner Profile (LLP)** from heterogeneous inputs, and use an LLM to *reason over, simulate, and interact with* that profile on behalf of the student.

The twin should be able to:

- Answer *"What topics is this student weak in?"*
- Predict *"Will this student struggle with Module 4 given their Module 2 performance?"*
- Generate *"A personalized explanation of gradient descent for this specific student"*
- Simulate *"How would this student answer this exam question?"*

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3. Input Data — What Defines the Twin

Collect or simulate the following student attributes:

Category	Examples
Academic	Grades per subject, attendance, assignment scores
Behavioral	Time spent on LMS, login frequency, resource access patterns
Cognitive	Learning style (visual/verbal), prior quiz performance, error patterns
Demographic	Year of study, branch, background (optional)
Self-reported	Goals, preferred study time, self-assessed confidence per topic

For a semester project, **simulated or anonymized data is perfectly fine.** Students can generate synthetic cohorts of 50–100 student profiles.

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4. System Architecture

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┌──────────────────────────────────────────────────────────┐
│                    DATA INGESTION                        │
│  Grades / LMS logs / Quizzes / Self-reports              │
└──────────────────────────────────────────────────────────┘
                     │
                     ▼
┌──────────────────────────────────────────────────────────┐
│          LIVE LEARNER PROFILE (LLP)                      │
│  Structured JSON / Text summarizing student state        │
│  Updated after each interaction or assessment            │
└──────────────────────────────────────────────────────────┘
                     │
          ┌──────────┴──────────┐
          ▼                     ▼
┌───────────────────┐   ┌───────────────────────┐
│  RETRIEVAL ENGINE │   │   LLM REASONING       │
│  Course material  │   │   Core twin engine    │
│  Past papers      │   │   (GPT/Gemini/Llama)  │
│  Rubrics          │   │                       │
└───────────────────┘   └───────────────────────┘
         │                        │
         └───────────┬────────────┘
                     ▼
┌──────────────────────────────────────────────────────────┐
│                   TWIN CAPABILITIES                      │
│  • Weakness diagnosis                                    │
│  • Personalized explanation generation                   │
│  • Performance prediction                                │
│  • Exam answer simulation                                │
│  • Study plan recommendation                             │
└──────────────────────────────────────────────────────────┘
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5. Step-by-Step Build Plan (10 Weeks)

Phase 1 — Data & Profile Construction (Weeks 1–3)

Week 1: Data Design

- Define the schema for the Live Learner Profile (LLP)

- Decide which attributes to include (start with 8–10 fields)
- Generate or collect a synthetic dataset of 50 student profiles
- Tools: Python, Faker library for synthetic data, Google Forms for real data

Week 2: Profile Builder

- Write a pipeline that converts raw student data (CSV/JSON) into a structured LLP
- The LLP should be both machine-readable (JSON) and human-readable (short paragraph summary)
- Example LLP summary:

  "Priya is a 2nd-year CSE student with strong performance in programming (avg 82%) but consistent weakness in Mathematics (avg 51%). She accesses the LMS primarily late at night, submits assignments on time, and has self-reported low confidence in probability topics."

Week 3: Profile Update Logic

- Build a simple rule-based or LLM-based mechanism to update the LLP after a new quiz or assignment
- Test with 5 synthetic update events per student

Phase 2 — Core Twin Capabilities (Weeks 4–7)

Week 4: Weakness Diagnosis

- Prompt the LLM with the LLP + course topic list
- Ask it to identify weak areas with reasoning
- Evaluate: does it match ground-truth weak topics in your synthetic data?

Week 5: Personalized Explanation Generation

- Given a concept (e.g., "Bayes' Theorem") and a student's LLP, generate an explanation calibrated to their background and weak areas
- Compare outputs across students with different profiles — the explanation should visibly differ

Week 6: Performance Prediction

- Frame as: *"Given this student's LLP, predict their likely score on the upcoming Module 3 quiz (High / Medium / Low)"*
- Run on 20 synthetic students, compare LLM predictions to ground-truth simulated scores
- Baseline: predict mean score for everyone

Week 7: Exam Answer Simulation

- Given a student's LLP + an exam question, ask the LLM to simulate *how that student would answer*
- This is the most research-adjacent task — tests whether the twin captures individual behavior

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Phase 3 — Interface & Evaluation (Weeks 8–10)

Week 8: Streamlit Interface

- Build a simple UI with two views:
  - **Teacher view**: See all student twins, get class-wide insights
  - **Student view**: See your own twin, get personalized recommendations

Week 9: Evaluation

- Define metrics for each capability:

Capability	Metric
Weakness Diagnosis	Precision/Recall vs. ground-truth weak topics
Explanation Quality	Human rating (1–5) by 3 raters
Performance Prediction	Accuracy / F1 over simulated ground truth
Answer Simulation	Similarity to expected answer profile

Week 10: Report & Demo

- Write a 6–8 page report
- Record a 5-minute demo video
- Prepare a poster if required

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6. Tech Stack

Component	Tool
LLM	Gemini API (free tier) or Groq (Llama 3, free)
Embeddings	`sentence-transformers` (free, local)
Vector Store	FAISS or ChromaDB
Backend	Python + LangChain or direct API calls
UI	Streamlit
Data	Synthetic via Faker, or anonymized real data
Evaluation	Custom scripts + Google Forms for human eval

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7. Extensions (if time permits)

- **Temporal twin**: track how the student's twin evolves week by week
- **Counterfactual reasoning**: *"If this student had attended 3 more lectures, would their score improve?"*
- **Privacy-preserving twin**: explore what happens when you redact sensitive attributes — does the twin degrade gracefully?
- **Multi-student comparison**: *"Which students are most similar to each other?"* — clustering over LLP embeddings

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8. Research Angle (for ambitious groups)

This project connects to active research in:

- **LLM-based user simulation** — can LLMs faithfully simulate individual human behavior?
- **Personalized education / adaptive learning systems**
- **AI in education (AIEd)**
- **Synthetic data generation for education**

A rigorous version of the **answer simulation** task — with proper evaluation — could be written up as a short paper targeting **EDM (Educational Data Mining)** or **L@S (Learning at Scale)** workshops.

Want me to generate the full 8-section lab-format spec (with baselines, risk assessment, and weekly milestones) for this, or create the synthetic data schema and starter code structure?